ATHAR HUSSAIN

AI Engineer | Machine Learning Engineer | Data Scientist (Fresher)

Hyderabad, India | +91 9550030812 | atharhussain2910@gmail.com | linkedin.com/in/athar-hussain-datascience
github.com/ATHAR-HUSSAIN-2910 | Portfolio: athar-hussain-2910.github.io

PROFESSIONAL SUMMARY

Final-year B.Tech CSE student (graduating 2027) with hands-on skills in Python, SQL, Machine Learning, Deep Learning, and Generative AI. Built **11 end-to-end projects** and deployed **5 live applications**, including **ShopAssist AI**, a LangGraph-based customer support agent with RAG, PostgreSQL order tools, and a **28-case evaluation suite with a 100% pass rate**. Seeking entry-level AI Engineer, ML Engineer, and Data Scientist roles.

TECHNICAL SKILLS

Programming Languages & Databases: Python (OOP), SQL (JOINs, CTEs, Window Functions), PostgreSQL, SQL Server
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Cleaning, Feature Engineering, Outlier Detection
Machine Learning: Scikit-learn, Linear/Logistic Regression, Ridge, Lasso, Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost, K-Means, Pipelines, Cross-Validation, GridSearchCV, Optuna, SHAP
Deep Learning: PyTorch, ANN, CNN, Dataset/DataLoader, Batch Normalization, Dropout, Early Stopping, Data Augmentation
Generative AI & Agentic AI: LLMs, RAG, LangGraph, LangChain, FAISS, Sentence Transformers, Groq API, Tool Calling
Model Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC, R^2, MAE, RMSE, Confusion Matrix, Threshold Optimization
Tools & Deployment: Streamlit, Streamlit Community Cloud, Git, GitHub, Jupyter Notebook, VS Code, Google Colab
Currently Learning: Advanced RAG | LLM Fine-tuning | MLOps | AI System Design

PROJECTS

ShopAssist AI: Agentic Customer Support Agent | Live Demo | GitHub
Tech: Python, LangGraph, LangChain, Groq LLM, RAG, FAISS, PostgreSQL, SQLAlchemy, Pydantic, Streamlit
- Built a stateful LangGraph agent where a Groq LLM classifies intent and conditional routing sends each request to RAG policy retrieval (FAISS), PostgreSQL order tools (tracking, cancellation), or human escalation.
- Implemented safety controls (business rules in Python rather than prompts, cancellation confirmation, RAG relevance threshold against hallucination, exponential-backoff retries) and persistent multi-turn memory via a PostgreSQL checkpointer.
- Validated the agent with a deterministic **28-case evaluation suite** across 8 categories (routing, memory, context leakage, escalation), achieving a **100% pass rate**; deployed on Streamlit Cloud.

Machine Learning RAG Assistant | Live Demo | GitHub
Tech: Python, RAG, FAISS, Sentence Transformers (all-MiniLM-L6-v2), Groq LLM, LangChain, Streamlit
- Built a conversational RAG app using Sentence Transformer embeddings and FAISS semantic retrieval to ground Groq LLM answers in an ML knowledge base; added follow-up question rewriting, source-topic display, and out-of-scope handling.

Diabetes Risk Prediction with ANN | Live Demo | GitHub
Tech: PyTorch, ANN, Optuna, Scikit-learn, Pandas, Streamlit
- Developed a feed-forward PyTorch ANN on **70,692 BRFSS records** with leakage-safe preprocessing and stratified 70/15/15 splits; tuned with Optuna, early stopping, and validation-based threshold optimization.
- Achieved **0.830 ROC-AUC, 89.4% recall, and 77.6% F1** on the untouched test set; deployed as a Streamlit app.

CIFAR-10 Image Classification with Custom CNN | GitHub
Tech: PyTorch, CNN, Torchvision, Optuna, Data Augmentation
- Designed a custom Conv-BatchNorm-ReLU-MaxPool CNN for 60,000 32x32 RGB images with training-only data augmentation and Optuna tuning; reached **77.22% test accuracy (0.772 macro F1)** with confusion-matrix error analysis.

Customer Churn Prediction | Live Demo | GitHub
Tech: Python, XGBoost, Scikit-learn, SHAP, Pandas, Streamlit
- Compared six classification models and optimized XGBoost with GridSearchCV, reaching **79.7% accuracy and 0.848 ROC-AUC**; applied SHAP for model explainability and deployed the model as a Streamlit app.

House Price Prediction | Live Demo | GitHub
Tech: Python, Scikit-learn, Pipeline, ColumnTransformer, Streamlit
- Built Linear, Ridge, Lasso, and Elastic Net regression pipelines with feature engineering, cross-validation, and hyperparameter tuning; achieved **R^2 = 0.823** with Lasso and deployed for real-time predictions.

Customer Order SQL Project: Database Design & Analytics | GitHub
Tech: SQL Server, SSMS, Python, Pandas, Faker
- Designed a relational e-commerce database (200 customers, 40 products, 1,484 orders), generated synthetic data with Python and Faker, and wrote **85 SQL queries** using JOINs, CTEs, window functions, and aggregations.

Additional Projects: Titanic Comparative Classification Study | Titanic Survival Prediction | Customer Segmentation (K-Means) | Retail Sales Data Analysis

EDUCATION

B.Tech in Computer Science Engineering | Shadan College of Engineering and Technology, Hyderabad | Expected 2027 | **SGPA: 9+** (6 semesters)

SOFT SKILLS

Analytical Thinking | **Problem Solving** | **Communication** | **Ownership** | **Adaptability** | **Attention to Detail**